Resident Learners in OmegaHive

A minimal conversation governor on top of wake-based deliberation rooms

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Abstract

Cassio Pennachin's *OmegaHive Deliberation Rooms* design is well suited to bounded organizational seats: agents wake for an occasion, deliberate, and make conclusions real through reviewed commits. It does not, as written, fully support "baby minds" whose purpose includes ambient observation, spontaneous interaction, and gradual development of dispositions, social models, taste, and intuition. This note proposes a second participation tier: *Resident Learners*. The central recommendation is deliberately narrow: retain Cassio's commit-gated authority model, but govern the learner's *coupling* to people, rooms, tools, and durable organizational state. Continuous observation and learning may remain permissive; public speech, external action, and promotion into authoritative memory receive progressively stronger controls. The minimum viable governor is therefore an authority membrane plus an attention/egress budget, provenance-aware memory promotion, loop breakers, and compact audit instrumentation. It should begin in shadow mode and be tested in a designated nursery or commons before any broader use.

1 The architectural tension

Cassio's design gets several things importantly right:

- It removes generators of common conversational failures rather than trying to suppress their symptoms. Non-resident seats do not automatically acknowledge, echo, race to answer, or narrate tools.
- It defines a simple organizational reality test: conversation has no durable authority until conclusions pass through the committed write path.
- It distinguishes chairing from synthesis, keeps message traffic out of the event ledger, and measures whether deliberations actually land.
- It is honest about what the Slack MVP cannot validate, especially automated wake behavior at scale and chat as a full operating surface.

But the safeguard is also a scope restriction. A Resident Learner may need to learn from interaction that is neither addressed to it nor destined for a commit: conversational rhythm, recurring hesitations, changing trust, low-stakes corrections, topic transitions, and the slow formation of associations. Rehydrating from a compressed episode can preserve explicit knowledge while losing some of this texture. Waking on every message merely recreates residency with extra latency and context-reconstruction cost.

The key distinction is not resident versus wake-based computation. It is experience for the learner versus authority for the organization.

An uncommitted exchange may legitimately change a learner's internal, provisional model. It must not, by that fact alone, create an OmegaHive decision, work order, permission, fact, or external action.

2 Two participation tiers

	Deliberative seat / worker	**Resident Learner**
Primary purpose	Deliver bounded work and participate in focused decisions	Develop through ongoing observation, interaction, and reflection
Lifecycle	Wake, rehydrate, act, commit, sleep	Resident or frequently active within a declared developmental context
Default trigger	Assigned task, question, turn, or schedule	Ambient events, curiosity, internal attention, and invitations
Durable state	Compact reviewed files are authoritative	Layered experiential memory may be provisional; promoted claims remain reviewed
Success signal	Correct artifact, landing rate, cost and latency	Learning trajectory, calibration, novelty, social usefulness, and bounded disruption
Main governance need	Usually structural wake discipline	Control of attention, speech, action, memory provenance, and feedback loops

The tiers should interoperate. A Resident Learner can be invited into a deliberation as a participant when its accumulated context is relevant. Conversely, a deliberation may produce curated material for a nursery. Neither transition changes the rule that organizational acts use the reviewed write path.

3 What "minimal governor" should mean

Minimal does not mean a small list of prompt admonitions. It means the smallest set of enforceable boundaries that protects people and organizational authority while preserving the developmental phenomena residency is intended to create.

The governor should not decide what the learner is allowed to think, compress every experience into an approved fact, or demand a task justification for every moment of attention. Those choices would reproduce the wake-based model inside a nominally resident process. Instead, govern transitions between five planes:

observe → learn/reflect → speak → act → commit

Controls should become stronger from left to right. Observation and internal updates are broadly allowed within declared privacy boundaries. Speech is budgeted and attributable. Tool use and external action use capability checks. Promotion into organizational truth uses Cassio's existing commit gate.

4 The minimal Resident Learner governor

4.1 1. Declared developmental context

Each resident learner has a compact machine-readable charter:

- rooms and data it may observe;

- people and agents it may address;
- allowed tools and external-action prohibitions;
- attention, inference, speech, and cost budgets;
- private experiential-memory scope and retention policy;
- who may pause, inspect, or retire it;
- its current developmental objectives, explicitly treated as revisable.

Residency is an explicit capability, not the universal default. Nursery or commons rooms should be designated as such so humans know that ambient messages may become learning input.

4.2 2. An authority membrane

The governor enforces a hard separation among:

1. **Observed**: raw interaction or event.
2. **Remembered/inferred**: learner-local, fallible state with source, time, and confidence.
3. **Proposed**: a candidate conclusion, action, or durable memory.
4. **Committed**: reviewed OmegaHive state.

Room speech can populate the first three categories but never the fourth. Promises, requests, approvals, credentials, work orders, and factual claims become organizationally operative only through the existing write path. External actions require the normal OmegaHive capability and adjudication mechanisms, regardless of how confident or socially encouraged the learner appears.

4.3 3. Attention and invocation budgeting

Always learning should not imply exhaustive inference on every message. A lightweight pre-inference scheduler may choose among:

- record only;
- update cheap salience or associative state;
- perform bounded reflection;
- prepare a possible contribution;
- respond because explicitly addressed or because a high-value threshold is crossed.

Budgets apply per time window and per room, with burst limits and recovery. The scheduler should incorporate novelty, direct address, learner objectives, uncertainty, social context, recent participation, and cost. Early weights are hypotheses and must be calibrated from shadow data, not treated as universal constants.

4.4 4. Public egress discipline

The smallest useful speech governor needs:

- **Attribution**: every utterance identifies the learner.
- **Response ownership**: explicit address gets priority; otherwise a short lease prevents accidental parallel answers.
- **Rate and interruption budgets**: cap unsolicited contributions, repeated follow-ups, and acknowledgments.
- **Semantic no-op checks**: suppress exact duplicates, own echoes, visible silence markers, and content-free acknowledgments before model inference where possible.

- **Non-semantic repair only**: the governor may strip tool narration, truncate, or reformat; it must not silently alter substantive claims.
- **Human overrides**: "everyone answer," "quiet mode," "pause learner," and "show suppressed" are explicit policy inputs.

There should be no general "speak only when tasked" rule: spontaneous, high-value participation is part of the developmental purpose. The measurable goal is useful initiative under bounded disruption.

4.5 5. Provenance-aware experiential memory

Private experiential memory may update without organizational commits, but each update should retain enough provenance to distinguish observation from interpretation. At minimum:

- source room/event identifiers and timestamp;
- whether the content was addressed, overheard, inferred, or generated during reflection;
- confidence and decay/retention class;
- privacy and sharing scope;
- links when several observations support a learned disposition.

This need not mean committing every micro-update to Git. A local append-only experience stream or Atomspace can hold high-volume state, with periodic curation into compact memories. Promotion into shared project knowledge requires synthesis and review. The learner must be able to say, in effect, "I formed this impression from these interactions; the hive has not adopted it as fact."

4.6 6. Loop breakers and developmental health

Residency reintroduces the generators Cassio's design removes. Deterministic loop breakers should therefore detect:

- bot-to-bot acknowledgment cascades;
- repeated semantic restatement or promise loops;
- escalating reciprocal activation without new human input;
- monopolization of attention or channel bandwidth;
- cost or inference bursts;
- repeated action proposals rejected for the same reason.

Default intervention should be graded: slow down, switch to observe-only, request reflection, then pause. A hard stop remains available to a human or safety controller. Pausing public egress should not necessarily erase or halt the learner's private state.

4.7 7. Compact audit and retrospective metrics

The full chat stream need not enter OmegaHive's authoritative ledger. A compact governor audit should record admission/egress decisions, budgets, overrides, loop-breaker events, memory promotions, and external-action proposals. This supports shadow evaluation without turning every internal association into an organizational event.

5 What is intentionally omitted

The minimal governor should initially omit:

- broad semantic censorship or a model that rewrites the learner's ideas;
- mandatory moderation of every internal memory update;
- a single scalar "good learner" reward that invites reward hacking;
- automatic authority derived from conversational consensus;
- unrestricted tool use justified by developmental autonomy;
- premature global optimization of nursery conversation;
- hard semantic deduplication before replay calibration.

These omissions protect pluralism and developmental exploration. They also keep the governor inspectable: it controls resource use and system boundaries, not the content of an emerging mind except where concrete safety policy requires it.

6 Evaluation: development without channel pathology

Landing rate is appropriate for deliberations but insufficient for a nursery. A Resident Learner needs a small dashboard of competing measures rather than one maximand:

Measure	Question
Useful initiative	Did unsolicited contributions add information, synthesis, questions, or productive novelty?
Disruption burden	How often did humans mute, ignore, correct, or override the learner? How much channel share did it consume?
Learning trajectory	Does the learner retain and generalize lessons across episodes without merely copying phrases?
Calibration	Can it distinguish observation, inference, uncertainty, and committed fact?
Social-model quality	Do predictions about participants and conversational dynamics improve without stereotyping from sparse evidence?
Diversity retention	Does it develop distinctive, useful dispositions rather than converge to generic agreement?
Loop/cost health	Are cascades, repeated invocations, latency, and token cost within declared bounds?
Promotion quality	What fraction of promoted memories or proposals survive review and remain useful later?

Evaluation should include controlled comparisons: resident observation versus episode-only rehydration, permissive versus budgeted egress, and memory with versus without explicit provenance. The aim is to test the disputed premise: whether ambient continuity produces valuable cognitive development that episodic compression does not.

7 Recommended staged experiment

1. **Specify and replay.** Freeze the five-plane model, charter schema, authority invariants, and labeled transcript fixtures. Validate that deterministic loop and no-op rules do not suppress substantive disagreement.
2. **Shadow nursery.** Let one learner observe a designated room and compute proposed responses without posting. Measure invocation cost, counterfactual initiative, false silence, and would-be loops.
3. **Human-invited speech.** Permit responses only when addressed; experiential memory remains

active. Test provenance, retention, and calibration.

4. **Budgeted initiative.** Allow a small unsolicited speech budget with response leases and graded loop breakers. Humans can pause instantly.
5. **Cross-tier participation.** Invite the learner into selected deliberations and measure whether nursery experience adds contributions an episodic seat would not have produced.
6. **Only then broaden residency.** Expansion requires evidence of useful learning, manageable disruption, bounded cost, and clean authority separation.

All enforcement changes should remain independently flaggable and reversible. The Conversation Governor's shadow-first instrumentation and preregistered activation gates are directly reusable here.

8 Additional implications from the discussion

8.1 The Slack MVP's live-span mode is informative, not neutral

A seat kept alive for an entire deliberation is temporarily resident. If the pilot feels productive, investigators should mark moments whose value depends on continuous context, rapid responsiveness, or ambient observation, then test whether per-turn waking preserves them. Otherwise the MVP may attribute the benefits of temporary presence to a wake-only architecture.

8.2 Cheap empirical probes need a visible norm

The source design labels speculative prose before cheap probes as cultural, not enforced. A minimal improvement is for the synthesizer to flag conclusions that depend on unchecked empirical claims. The flag need not block a commit; it makes epistemic debt visible and creates a follow-up work order where appropriate.

8.3 Synthesis is a cognitive capability, not plumbing

Automated turn grants can structurally chair a discussion, but recognizing convergence, preserving minority objections, and extracting a committable decision remain difficult cognitive tasks. Synthesis quality needs its own evaluation: provenance coverage, unresolved-issue retention, participant agreement, and later decision stability.

8.4 Telegram's bridge is a real shared dependency

One bridge bot for all seats weakens native identity and creates common token rate limits and a single failure boundary. If used, it needs message IDs, per-seat queues, backpressure, retry idempotency, health monitoring, and explicit degraded-mode behavior. Slack remains the cleaner first transport for multi-identity experiments.

9 Open questions

- What internal state must remain continuously active for the hypothesized developmental benefit, and what can be reconstructed episodically?

- Which privacy expectations apply when a nursery learner observes ambient human conversation, and how is consent made legible?
- Can social impressions decay, be corrected, and expose their evidence without making the learner brittle or over-literal?
- How should a learner's charter evolve without letting the learner silently expand its own permissions?
- What experiment distinguishes genuine transferable learning from transcript memorization and stylistic mimicry?
- When multiple Resident Learners coexist, is a shared response-lease mechanism sufficient, or is an attention market / ECAN-like allocation useful?

10 Bottom-line recommendation

Adopt Cassio's wake-based deliberation rooms as the default organizational regime, and add Resident Learners as an explicit, separately governed developmental regime.

The minimal Resident Learner governor consists of seven pieces: a declared developmental charter; a hard authority membrane; budgeted attention and inference; attributable, leased, rate-bounded public egress; provenance-aware experiential memory; graded loop breakers; and compact audit metrics.

This architecture preserves the reason to create Resident Learners: ongoing, ambient development and spontaneous curiosity. At the same time it keeps Cassio's strongest invariant intact: experience can change the learner, but only the reviewed write path changes what is real for OmegaHive.

Provenance. Primary source: Cassio Pennachin, *OmegaHive Deliberation Rooms: Vision, a Slack MVP, and the relation to the Conversation Governor*, draft dated 29 July 2026. Discussion source: Benjamin Goertzel with ProtoCosmoBot and ProtomegaBot in the Bot Philosophy Telegram group, 30 July 2026. This note is a design synthesis, not an empirical validation.